Basin MCMC Assignment 1

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The purpose of this assignment is to break down the development of the initial basin probability distribution and probability in the Basin MCMC algorithm.

Question 1. **[pts.]**

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\top \in \mathcal{X}_1$. Provide a function that approximates the log posterior $\log p(\theta \mid y)$ on $\mathcal{X}_1$, where the observed data is the noise-free data $y = A(\theta_0)$ generated by this atom.

The function $f_1 : \mathcal{X}_1 \to \mathbb{R}$ should satisfy the following properties:

1. unimodal;
2. a good approximation to the log posterior in the region of high probability around θ_0;
3. easy to compute defining quantities from θ_0;
4. easy to sample from;
5. as simple as possible.

Note that this function need only be specified up to an additive constant.

Solution. We present the solution as follows:

1. Define f_1 as the second-order Taylor polynomial of the log posterior about θ_0;
2. Derive a formula for the gradient $g = \nabla \log p(\theta_0 \mid y)$;
3. Derive a formula for the negative Hessian $P = -\nabla^2 \log p(\theta_0 \mid y)$;
4. State the final formula for f_1.

1. For $\theta \in \mathcal{X}_1$, the log posterior is given by

$$\log p(\theta \mid y) = -\frac{1}{2\sigma^2}\|y - A(\theta)\|_2^2 + \log p(\theta) + C,$$

where C is some constant independent of θ, and $A : \mathcal{X} \to [0,1]^{N_t N_x}$ is the forward map. Performing a second-order Taylor expansion about θ_0 (since $\log p(\theta \mid y) \in C^3(U)$ for some open neighbourhood $U \subset \mathcal{X}_1$ of θ_0, provided $\theta_0 \in \operatorname{int}(\mathcal{X}_1)$) gives

$$\log p(\theta \mid y) = \log p(\theta_0 \mid y) + g^\top (\theta - \theta_0) - \frac{1}{2} (\theta - \theta_0)^\top P (\theta - \theta_0) + O(\|\theta - \theta_0\|^3), \tag{1}$$

where

$$g := \nabla \log p(\theta_0 \mid y) \quad \text{and} \quad P := -\nabla^2 \log p(\theta_0 \mid y).$$

Motivated by (1), we define our approximation to the log posterior $f_1 : \mathcal{X}_1 \to \mathbb{R}$ by

$$f_1(\theta) := \log p(\theta_0 \mid y) + g^\top (\theta - \theta_0) - \frac{1}{2} (\theta - \theta_0)^\top P (\theta - \theta_0).$$

2. We have that

$$g = \nabla \log p(\theta_0 \mid y) = \nabla \left(\log p(y \mid \theta_0) + \log p(\theta_0) - \log p(y)\right) = \nabla \log p(y \mid \theta_0) + \nabla \log p(\theta_0). \tag{2}$$

For $\theta \in \mathcal{X}_1$, we have that

$$\nabla \log p(y \mid \theta) = \nabla \left(-\frac{1}{2\sigma^2}\|y - A(\theta)\|_2^2\right) = \frac{1}{\sigma^2} J_A(\theta)^\mathsf{T} \left(y - A(\theta)\right),$$

where $J_A(\theta)$ is the Jacobian of the forward map A (restricted to the domain $\mathcal{X}_1$). Recalling that $y = A(\theta_0)$,

$$\nabla \log p(y \mid \theta_0) = \frac{1}{\sigma^2} J_A(\theta_0)^\mathsf{T} \left(y - A(\theta_0)\right) = 0. \tag{3}$$

Finally, for $\theta \in \mathcal{X}_1$, we have that

$$\nabla \log p(\theta) = \nabla \log \left(\lambda_R \mathrm{e}^{-\lambda_r R}\right) = \nabla \left(-\lambda_R R\right) = \left(0, 0, -\lambda_R, 0\right)^\mathsf{T}. \tag{4}$$

Combining (2), (3), and (4), we have that

$$g = \left(0, 0, -\lambda_R, 0\right)^\mathsf{T}.$$

3. The negative Hessian of the log posterior is given by

$$P = -\nabla^2 \log p(\theta_0 \mid y) = -\nabla^2 \log p(y \mid \theta_0) - \nabla^2 \log p(\theta_0). \tag{5}$$

Recalling (4), we have that

$$\nabla^2 \log p(\theta_0) = 0. \tag{6}$$

Next, for $\theta \in \mathcal{X}_1$, one can show that

$$\nabla^2 \log p(y \mid \theta) = -\frac{1}{\sigma^2} J_A(\theta)^\mathsf{T} J_A(\theta) + \frac{1}{\sigma^2} \sum_k (y - A(\theta))_k \nabla^2 A_k(\theta).$$

Again recalling that $y = A(\theta_0)$, we obtain

$$\nabla^2 \log p(y \mid \theta_0) = -\frac{1}{\sigma^2} J_A(\theta_0)^\mathsf{T} J_A(\theta_0). \tag{7}$$

Combining (5), (6), and (7), we have that

$$P = \frac{1}{\sigma^2} J_A(\theta_0)^\mathsf{T} J_A(\theta_0). \tag{8}$$

To obtain a formula for P, we will obtain a formula for $J_A(\theta_0)$. To this end, recall that the $tN_x + x$-th entry of the forward map $A : \mathcal{X} \to [0,1]^{N_t N_x}$ with parameters $\theta = (x_0, d, R, a)^\mathsf{T} \in \mathcal{X}_1$ is given by

$$A(t, x; \theta) = aB(x)w(t - T(x))$$

where

$$B(x) = \sqrt{\frac{d+R}{s(x)}} e^{-2\alpha s(x)}, \quad T(x) = \frac{2}{v}(s(x) - (d+R)) + \frac{2d}{v} = \frac{2}{v}(s(x) - R)$$

with $s(x) = \sqrt{(d+R)^2 + (x - x_0)^2}$, and

$$w(\tau) = (1 - 2(\pi f_0 \tau)^2) e^{-(\pi f_0 \tau)^2}.$$

Applying the product and chain rules, we obtain

$$
\begin{aligned}
\frac{\partial A(t,x;\theta)}{\partial x_0} &= aB(x)\left[w(t-T(x))\frac{\partial \log B(x)}{\partial x_0} + w'(t-T(x))\frac{-\partial T(x)}{\partial x_0}\right], \\
\frac{\partial A(t,x;\theta)}{\partial d} &= aB(x)\left[w(t-T(x))\frac{\partial \log B(x)}{\partial (d+R)} + w'(t-T(x))\frac{-\partial T(x)}{\partial d}\right], \\
\frac{\partial A(t,x;\theta)}{\partial R} &= aB(x)\left[w(t-T(x))\frac{\partial \log B(x)}{\partial (d+R)} + w'(t-T(x))\frac{-\partial T(x)}{\partial R}\right], \\
\frac{\partial A(t,x;\theta)}{\partial a} &= B(x)\, w(t-T(x)).
\end{aligned} \tag{9}
$$

The needed derivatives are given by

$$
\begin{aligned}
w'(\tau) &= 2(\pi f_0)^2\tau(2(\pi f_0\tau)^2-3)e^{-(\pi f_0 \tau)^2}, \\
\frac{\partial \log B(x)}{\partial x_0} &= (x-x_0)\left(\frac{1}{2s(x)^2}+\frac{2\alpha}{s(x)}\right), \\
\frac{\partial \log B(x)}{\partial (d+R)} &= \frac{1}{2(d+R)} - \frac{(d+R)}{2s(x)^2} - \frac{2\alpha(d+R)}{s(x)}, \\
-\frac{\partial T(x)}{\partial x_0} &= \frac{2(x-x_0)}{vs(x)}, \\
-\frac{\partial T(x)}{\partial d} &= -\frac{2(d+R)}{vs(x)}, \\
-\frac{\partial T(x)}{\partial R} &= \frac{2}{v}\left(1-\frac{(d+R)}{s(x)}\right).
\end{aligned}
$$

Vectorising each of the derivatives in (9) gives the four columns of $J_A(\theta_0)$.

4. Recall that for $\theta \in \mathcal{X}_1$, our approximation to the log posterior is given by

$$
f_1(\theta) = \log p(\theta_0 \mid y) + g^{\mathsf{T}}(\theta-\theta_0) - \frac{1}{2}(\theta-\theta_0)^{\mathsf{T}} P (\theta-\theta_0).
$$

Assuming that P is positive definite and so P^{-1} exists (which holds if and only if $J_A(\theta_0)$ has full rank), we may complete the square to obtain

$$
\begin{aligned}
f_1(\theta) &= \log p(\theta_0 \mid y) + g^{\mathsf{T}}(\theta-\theta_0) - \frac{1}{2}(\theta-\theta_0)^{\mathsf{T}} P (\theta-\theta_0) \\
&= \log p(\theta_0 \mid y) - \frac{1}{2}\left[(\theta-\theta_0)^\top P(\theta-\theta_0) - 2g^\top(\theta-\theta_0)\right] \\
&= \log p(\theta_0 \mid y) - \frac{1}{2}\left[(\theta-\theta_0)^\top P(\theta-\theta_0) - 2g^\top(\theta-\theta_0) + g^\top P^{-1}g - g^\top P^{-1}g\right] \\
&= \log p(\theta_0 \mid y) - \frac{1}{2}\left[(\theta-\theta_0-P^{-1}g)^\top P(\theta-\theta_0-P^{-1}g) - g^\top P^{-1}g\right] \\
&= \log p(\theta_0 \mid y) + \frac{1}{2}g^\top P^{-1}g - \frac{1}{2}(\theta-\theta_0-P^{-1}g)^\top P(\theta-\theta_0-P^{-1}g) \\
&= C' - \frac{1}{2}(\theta-\mu)^\top P(\theta-\mu),
\end{aligned}
$$

where C' is some constant independent of θ, and $\mu := \theta_0 + P^{-1}g$.

Thus, our approximation $f_1(\theta)$ defines a truncated Gaussian on $\mathcal{X}_1$, with mean μ and covariance P^{-1}. In the implementation, the Moore–Penrose pseudoinverse P^+ is used in place of P^{-1} for numerical robustness.

Question 2. **[pts.]**

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\intercal \in \mathcal{X}_1$. Provide a function that approximates the log likelihood $\log p(y \mid \theta)$ on $\mathcal{X}_2$, where the observed data is the noise-free data $y = A(\theta_0)$ generated by this atom.

The function $g_2 : \mathcal{X}_2 \to \mathbb{R}$ should satisfy the following properties:

1. unimodal;
2. exchangeable, i.e., $g_2(\theta_1, \theta_2) = g_2(\theta_2, \theta_1)$;
3. a good approximation to the log likelihood in the region of high probability around θ_0;
4. easy to compute defining quantities from θ_0;
5. easy to sample from;
6. as simple as possible.

Note that this function need only be specified up to an additive constant.

Solution. TODO: read over. Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\intercal \in \mathcal{X}_1$. We want to approximate the log posterior on $\mathcal{X}_2$, where the observed data is the noise-free data $y = A(\theta_0)$ generated by this atom.

For simplicity, we will start by approximating the log likelihood on $\mathcal{X}_2$, where the observed data is the noise-free data $y = A(\theta_0)$ generated by this atom.

This is given by

$$\log p(y \mid \theta_1, \theta_2) = -\frac{1}{2\sigma^2}\|A(\theta_0) - A(\theta_1) - A(\theta_2)\|_2^2 + C$$

for some constant C. Recall that the forward map is given by

$$A(t, x; \theta) = aB(x)w(t - T(x))$$

where

$$B(x) = \sqrt{\frac{d+R}{s(x)}}e^{-2\alpha s(x)}, \quad T(x) = \frac{2}{v}(s(x) - (d+R)) + \frac{2d}{v} = \frac{2}{v}(s(x) - R)$$

with $s(x) = \sqrt{(d+R)^2 + (x - x_0)^2}$, and

$$w(\tau) = (1 - 2(\pi f_0 \tau)^2)e^{-(\pi f_0 \tau)^2}.$$

Reparameterization Define $g_0 := (x_0, d_0, R_0)^\intercal$, so that we may write the representative atom as

$$\theta_0 = (g_0, a_0)^\intercal.$$

Now, with two atoms

$$\theta_1 = (g_1, a_1)^\intercal \quad \text{and} \quad \theta_2 = (g_2, a_2)^\intercal,$$

where $g_1 := (x_1, d_1, R_1)^\intercal$ and $g_2 := (x_2, d_2, R_2)^\intercal$, we will use the following reparameterization:

- $m := a_1 + a_2$ is the total amplitude,
- $s := \frac{a_1}{a_1 + a_2}$ is the amplitude split,

- $c := sg_1 + (1-s)g_2$ is the amplitude-weighted center,
- $h := g_1 - g_2$ is the difference.

We may recover the original parameters as follows:

- $a_1 = sm$,
- $a_2 = (1-s)m$,
- $g_1 = c + (1-s)h$,
- $g_2 = c - sh$.

Rewrite the forward map Define the function $F : [0, X_{\max}] \times [0, D_{\max}] \times [0, \infty) \to [0,1]^{N_t N_x}$ by

$$F(g) := A(g, 1),$$

where $A(g,1)$ is the $N_t N_x$-dimensional vector with $A(t, x; g, 1)$ as the $tN_x + x$-th entry. Notice that we have that

$$A(g, a) = aF(g).$$

Hence,

$$\begin{aligned} A(\theta_1) + A(\theta_2) &= a_1 F(g_1) + a_2 F(g_2) \\ &= m\left[sF(g_1) + (1-s)F(g_2)\right] \\ &= m\left[sF(c+\delta_1) + (1-s)F(c+\delta_2)\right]. \end{aligned} \tag{10}$$

where $\delta_1 := g_1 - c = (1-s)h$ and $\delta_2 := g_2 - c = -sh$.

Taylor approximation to F around c Now, (since F is smooth) performing a second-order Taylor expansion of F around c gives

$$F(c+\delta) = F(c) + J_c\delta + \frac{1}{2}H_c[\delta, \delta] + O(\|\delta\|^3), \tag{11}$$

where J_c is the Jacobian of F at c, given by

$$J_c = \left[\frac{\partial F(c)}{\partial x_0}, \frac{\partial F(c)}{\partial d}, \frac{\partial F(c)}{\partial R}\right]$$

and $H_c[\delta, \delta]$ is the $N_t N_x$-dimensional vector with k-th entry

$$\left(H_c[\delta, \delta]\right)_k = \delta^\mathsf{T} \nabla^2 F_k(c)\delta,$$

where $F_k(c)$ is the k-th entry of $F(c)$.

Using the Taylor approximation (11) for each term in (10) gives

$$\begin{aligned} sF(c+\delta_1) + (1-s)F(c+\delta_2) = F(c) + J_c\left(s\delta_1 + (1-s)\delta_2\right) + \frac{1}{2}\left(sH_c[\delta_1, \delta_1] + (1-s)H_c[\delta_2, \delta_2]\right) \\ + O(\|\delta_1\|^3 + \|\delta_2\|^3). \end{aligned} \tag{12}$$

Now, notice that since $s\delta_1 + (1-s)\delta_2 = 0$, we have that

$$J_c\left(s\delta_1 + (1-s)\delta_2\right) = 0. \tag{13}$$

Next, notice that since $H_c[\delta, \delta]$ is bilinear, we have that

$$sH_c[\delta_1, \delta_1] + (1-s)H_c[\delta_2, \delta_2] = s(1-s)^2 H_c[h, h] + (1-s)s^2 H_c[h, h]$$

$$= s(1-s)H_c[h,h]. \tag{14}$$

Combining (12), (13), and (14) gives

$$sF(c+\delta_1) + (1-s)F(c+\delta_2) = F(c) + \frac{1}{2}s(1-s)H_c[h,h] + O(\|\delta_1\|^3 + \|\delta_2\|^3). \tag{15}$$

Since

$$\|\delta_1\|^3 + \|\delta_2\|^3 = \left(|s|^3 + |1-s|^3\right)\|h\|^3,$$

if we restrict to the region where $|a_1 + a_2| > \epsilon$ for some $\epsilon > 0$ (which should be reasonable otherwise they cancel each other out in the image), so that $|s|, |1-s| < 1/\epsilon$, then we have that

$$O(\|\delta_1\|^3 + \|\delta_2\|^3) = O\left(\|h\|^3\right). \tag{16}$$

Overall, combining (15) and (16) gives

$$sF(c+\delta_1) + (1-s)F(c+\delta_2) = F(c) + \frac{1}{2}s(1-s)H_c[h,h] + O\left(\|h\|^3\right) \tag{17}$$

when $|a_1 + a_2| > \epsilon$.

Taylor approximation to F around g_0 Now, (since F is smooth) performing a first-order Taylor expansion of F around g_0 gives

$$F(g_0 + \delta) = F(g_0) + J_{g_0}\delta + O(\|\delta\|^2), \tag{18}$$

where and J_{g_0} is the Jacobian of F at g_0, given by

$$J_{g_0} = \left[\frac{\partial F(g_0)}{\partial x_0}, \frac{\partial F(g_0)}{\partial d}, \frac{\partial F(g_0)}{\partial R}\right].$$

Applying the approximation (18) to $F(c)$ in (17) gives

$$\begin{aligned} sF(c+\delta_1) + (1-s)F(c+\delta_2) &= F(c) + \frac{1}{2}s(1-s)H_c[h,h] + O\left(\|h\|^3\right) \\ &= F(g_0 + q) + \frac{1}{2}s(1-s)H_c[h,h] + O\left(\|h\|^3\right) \\ &= F(g_0) + J_{g_0}q + \frac{1}{2}s(1-s)H_c[h,h] + O\left(\|q\|^2 + \|h\|^3\right). \end{aligned} \tag{19}$$

when $|a_1 + a_2| > \epsilon$, where $q := c - g_0$.

Approximate $H_c[h,h]$ by $H_{g_0}[h,h]$ Since we would like an approximation that only depends on the representative atom (g_0, a_0), we will approximate $H_c[h,h]$. Given some assumptions (such as F has bounded third derivatives in a neighbourhood of g_0), we have that

$$H_c[h,h] = H_{g_0}[h,h] + O(\|q\|\|h\|^2).$$

Combining this with (19) gives

$$\begin{aligned} sF(c+\delta_1) + (1-s)F(c+\delta_2) = F(g_0) + J_{g_0}q + \frac{1}{2}s(1-s)H_{g_0}[h,h] \\ + O\left(\|q\|^2 + \|q\|\|h\|^2 + \|h\|^3\right), \end{aligned} \tag{20}$$

when $|a_1 + a_2| > \epsilon$.

Approximation of the forward map Define $\eta := m - a_0$ to be the difference between the total amplitude and the amplitude of the representative atom. Then, from (20) and (10) we have that

$$A(\theta_1) + A(\theta_2) = m\left[sF(c+\delta_1) + (1-s)F(c+\delta_2)\right]$$

$$
\begin{aligned}
&= (a_0 + \eta)\left[sF(c+\delta_1) + (1-s)F(c+\delta_2)\right] \\
&= (a_0 + \eta)\big[F(g_0) + J_{g_0} q + \frac{1}{2} s(1-s) H_{g_0}[h,h] \\
&+ O\left(\|q\|^2 + \|q\|\|h\|^2 + \|h\|^3\right)\big] \\
&= a_0 F(g_0) + \eta F(g_0) + a_0 J_{g_0} q + \frac{a_0}{2} s(1-s) H_{g_0}[h,h] \\
&+ O\left(|\eta|\,\|q\| + |\eta|\,\|h\|^2 + \|q\|^2 + \|q\|\|h\|^2 + \|h\|^3\right) \\
&= A(\theta_0) + \eta F(g_0) + a_0 J_{g_0} q + \frac{a_0}{2} s(1-s) H_{g_0}[h,h] \\
&+ O\left(|\eta|\,\|q\| + |\eta|\,\|h\|^2 + \|q\|^2 + \|q\|\|h\|^2 + \|h\|^3\right)
\end{aligned}
$$

when $|a_1 + a_2| > \epsilon$. That is,

$$
\begin{aligned}
A(\theta_1) + A(\theta_2) - A(\theta_0) &= \eta F(g_0) + a_0 J_{g_0} q + \frac{a_0}{2} s(1-s) H_{g_0}[h,h] \\
&+ O\left(|\eta|\,\|q\| + |\eta|\,\|h\|^2 + \|q\|^2 + \|q\|\|h\|^2 + \|h\|^3\right)
\end{aligned} \tag{21}
$$

in some neighborhood around $\eta = m - a_0 = 0$.

Question 3. **[pts.]**

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\mathsf{T} \in \mathcal{X}_1$. Provide a function that approximates the log posterior $\log p(\theta \mid y)$ on $\mathcal{X}_2$, where the observed data is the noise-free data $y = A(\theta_0)$ generated by this atom.

The function $f_2 : \mathcal{X}_2 \to \mathbb{R}$ should satisfy the following properties:

1. unimodal;
2. exchangeable, i.e., $f_2(\theta_1, \theta_2) = f_2(\theta_2, \theta_1)$;
3. a good approximation to the log posterior in the region of high probability around θ_0;
4. easy to compute defining quantities from θ_0;
5. easy to sample from;
6. as simple as possible.

Note that this function need only be specified up to an additive constant. This should build on the function found in Question 2.

Solution.

Question 4. **[pts.]**

Fix $n \geq 2$. Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\mathsf{T} \in \mathcal{X}_1$. Provide a function that approximates the log posterior $\log p(\theta \mid y)$ on $\mathcal{X}_n$, where the observed data is the noise-free data $y = A(\theta_0)$ generated by this atom.

The function $f_n : \mathcal{X}_n \to \mathbb{R}$ should satisfy the following properties:

1. unimodal;
2. exchangeable;
3. a good approximation to the log posterior in the region of high probability around θ_0;
4. easy to compute defining quantities from θ_0;
5. easy to sample from;
6. as simple as possible.

Note that this function need only be specified up to an additive constant, and this constant need not be common to all n. This should be a generalization of the function found in Question 3.

Solution.

Question 5. **[pts.]**

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\mathsf{T} \in \mathcal{X}_1$. Rewrite the function found in Question 1 in the same form as the functions found in Question 4.

If it is not possible to write the function found in Question 1 in the same form as the functions found in Question 4, explain why. Then, provide an alternative function that approximates the log posterior $\log p(\theta \mid y)$ on $\mathcal{X}_1$ that is in the same form as the functions found in Question 4.

Solution.

Question 6. **[pts.]**

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\mathsf{T} \in \mathcal{X}_1$. Provide an approximation of the sequence of probabilities $\{p_n\}_{n\in\mathbb{N}}$ where

$$p_n := P(\theta \in X_n \mid y) = \int_{\mathcal{X}_n} p(\theta \mid y)\mathrm{d}\theta,$$

where $y = A(\theta_0)$ is the noise-free data generated by θ_0.

You may find it useful to use the functions found in Questions 4 and 5. Note that you need only approximate the sequence up to a common multiplicative constant; that is, it suffices to approximate quantities proportional to $\{p_n\}_{n\in\mathbb{N}}$.

Solution.

Question 7. **[pts.]**

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\mathsf{T} \in \mathcal{X}_1$. Provide an approximation of the probability p where

$$p :=,$$

where $y = A(\theta_0)$ is the noise-free data generated by θ_0.

Solution.